

30 At time t , a sample of a radioactive substance contains N atoms of a particular nuclide.

At time $(t + \Delta t)$, where Δt is a short period of time, the number of atoms of the nuclide is $(N - \Delta N)$.

Which of the following expressions gives the decay constant of the nuclide?

A $\frac{\Delta N}{N}$

B $\frac{N - \Delta N}{t + \Delta t}$

C $\frac{\Delta N}{N\Delta t}$

D $\frac{N\Delta N}{\Delta t}$

Decay constant is the fraction of the total number of undecayed nuclei present which decays per unit time. Suppose N is the size of a population of radioactive atoms at a given time t , and dN is the amount by which the population decreases in time dt ; then the rate of change is given by the equation $\frac{dN}{dt} = -\lambda N$, where λ is the decay constant.

$$\frac{dN}{dt} = -\lambda N \rightarrow \lambda = -\frac{1}{N} \frac{dN}{dt} = \frac{\Delta N}{N\Delta t}$$

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1	2	3	4	5	6	7	8	9	10
C	D	C	A	C	A	B	A	C	B
11	12	13	14	15	16	17	18	19	20
A	A	A	C	D	B	A	A	D	B
21	22	23	24	25	26	27	28	29	30
C	B	C	A	D	C	C	C	B	C